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MILITARY SPECIFICATION

TEST STAND, AIRCRAFT HYDRAULIC SYSTEM, TRAILER-MOUNTED, A/M27T-2, ELECTRIC-MOTOR-DRIVEN, TRIPLE SYSTEM

1. SCOPE

1.1 This specification covers a triple-system (each system providing 20 gpm at 3,000 psi and 10 gpm at 5,000 psi), electric-motor-driven, trailer-mounted, aircraft hydraulic system test stand, designated A/M27T-2.

2. APPLICABLE DOCUMENTS

* 2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

CC-M-641	Motor, Alternating Current, (Integral-Horsepower, 200 HP and Smaller)
GG-G-76	Gages, Pressure and Vacuum, Dial Indicating (For Air, Steam, Oil, Water, Ammonia and Chloro-Fluoro Hydrocarbon Gases)
PPP-B-621	Box, Wood, Nailed and Lock-Corner
PPP-T-60	Tape, Pressure Sensitive Adhesive, Waterproof, for Packaging

Military

MIL-P-116	Preservation, Methods of
MIL-B-121	Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-F-3541	Fittings, Lubrication
MIL-C-3600	Compressors, Air and Gas (Except Oxygen) Packaging of
MIL-H-5440	Hydraulic Systems, Aircraft Types I and II, Design, Installation, and Data Requirements for
MIL-R-5220	Reservoirs, Aircraft, Hydraulic, Non-Separated Type
MIL-H-5593	Hose, Aircraft, Low-Pressure, Flexible
MIL-H-5606	Hydraulic Fluid, Petroleum Base, Aircraft, Missile, and Ordnance
MIL-C-5756	Cable and Wire, Power, Electric, Portable
MIL-H-6083	Hydraulic Fluid, Petroleum Base, for Preservation and Testing
MIL-M-8090	Mobility, Towed Aerospace Ground Equipment, General Requirements for
MIL-A-8421	Air Transportability Requirements, General Specification for

FSC 4920

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MIL-H-8788	Hose, Hydraulic and Pneumatic, High Pressure
MIL-E-16298	Electric Machines Having Rotating Parts and Associated Repair Parts, Packaging of
MIL-F-27656	Filter, Fluid, Pressure, Absolute 5 Micron, Hydraulic
MIL-R-38489	Reels, Static Discharge, Grounding RMU-17/E and RMU-18/E

STANDARDS

Military

MIL-STD-100	Engineering Drawing Practices
MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-281	Automobiles, Trucks, Truck-Tractors, Trailers and Trailer Dollies, Preservation and Packaging of
MIL-STD-808	Finishes, Protective, and Codes, for Finishing Schemes for Ground and Ground Support Equipment
MIL-STD-810	Environmental Test Methods for Aerospace and Ground Equipment
MIL-STD-826	Electromagnetic Interference Test Requirements and Test Methods
MIL-STD-831	Test Reports, Preparation of
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing, With Appropriate Test Methods
MS27265	Connector, Plug, Electrical-Ramp Power, 416/240 Volts, Male
MS28720	Filter, Aircraft Hydraulic Line Type, 3,000 PSI
MS28741	Hose Assembly, Detachable End Fitting, Medium Pressure
MS28759	Hose Assembly, Rubber, Hydraulic and Pneumatic (3,000 PSI) Flared Tube
MS33656	Fitting End, Standard Dimensions for Flared Tube Connection and Gasket Seal
MS51335	Pintle Assembly, Towing-18,000 Pounds Capacity, Manual Release

Air Force-Navy Aeronautical

AN832	Union, Flared Tube 3/8 Bulkhead and Universal
AN924	Nut-Flared Tube, Bulkhead and Universal Fitting
AN929	Cap Assembly, Pressure Seal Flared Tube Fitting
AN6235	Filter Element-Hydraulic Replaceable Micronic Line Type
AN6236	Filter Element-Hydraulic Replaceable Micronic Reservoir Type
AN6249	Valve-3,000 PSI Hydraulic Check
AN6270	Hose Assembly - Detachable Swivel Fitting, Low Pressure
AND10050	Bosses, Standard Dimensions for Gasket Seal Straight Thread
AND10266	Flange-Type XVI Accessory Mounting
AND20006	Drive - Type XVI Engine Accessory

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Air Force

48B7796	Ring Assembly - Tie Down 10,000 Lb
49B6564	Ring - Eyebolt 25,000 Lb Cargo Tiedown

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Board of Fire Underwriters'

Pamphlet 70 National Electrical Code (Electrical Wiring and Apparatus)

(Application for copies should be addressed to the National Fire Protection Association, 60 Batterymarch Street, Boston, Massachusetts 02109.)

National Electrical Manufacturers Association

Standard MG-1 Motors and Generators

(Application for copies should be addressed to the National Electrical Manufacturers Association, 155 East 44th Street, New York, New York 10017.)

American Standards Association

Standard C-50 Rotating Electrical Machinery

(Application for copies should be addressed to the American Standards Association, 10 East 40th Street, New York, New York 10017.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 Components. The test stand shall consist of the following major components:

Item or System Name	See Requirement
Hydraulic systems	3.8
Electrical system	3.9
Pneumatic pressurizing system	3.10
Fan(s) or blower(s)	3.11
Trailer	3.12
Control panel	3.13
Housing	3.14

3.3 General specifications. The requirements of the following specifications, as specified, apply as requirements of this specification with the exceptions and additions specified herein. When the general specifications and this specification conflict, this specification shall govern.

a. MIL-H-5440 Requirements for materials, fluids, reservoir venting, tubing, and cleaning of parts and systems

b. MIL-M-8090 General requirements and those specified for type II, group C mobility.

3.4 Selection of specifications and standards. Specifications and standards for necessary commodities and services not

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specified herein shall be selected in accordance with MIL-STD-143.

3.4.1 Fittings. All fittings and threaded bosses in all fluid circuit components shall be as specified on AND10050 or MS33656. All 5,000-psi pressure fittings shall be steel. Bulkhead-type, high-pressure fittings shall use a locknut with a backup gasket.

3.5 Design. The test stand shall be a design with three independent systems, self-contained, mobile testing unit providing not less than 60 gpm total flow at 3,000 psi with the components arranged as shown on figure 1. The components shall be arranged for maximum ease of operation and maintenance consistent with the versatility and efficiency required to conform to the requirements specified herein. The test stand shall also be designed to provide 30 gpm total flow at 5,000 psi.

3.5.1 Reliability. The test stand shall have a minimum MTBF (mean-time-between-failures) of 152 hours with a minimum reliability of 0.967 for a 5-hour mission at a confidence factor of 0.90.

3.6 Construction. The test stand shall withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

3.6.1 Maintainability. The test stand shall be designed and constructed as specified herein and to provide the following:

a. Minimum number of parts consistent with the reliability and performance specified herein

b. Minimum amount of training and time necessary for assembly, disassembly, location of trouble sources, and maintenance including servicing. Where practical, parts and components shall be located or positioned for rapid and simple inspection and recognition of excessive wear or potential failure

c. Permit adjustments, servicing, replacement of parts and components, and other maintenance with minimum disturbance to other equipment parts or components. Parts and components shall be located for ample and rapid access unless performance or reliability will be appreciably degraded by the accessible location. In performing maintenance, if engineering reasoning or data determines that physical or visual interference between items cannot be avoided, the item predicted to require the most maintenance shall be located for best accessibility

d. Permit maintenance with general-purpose tools and equipment normally available commercially. Use of special-purpose tools and equipment shall be subject to approval by the procuring activity

e. Minimum number of tools required for maintenance by design practices such as reducing the variety of bolthead sizes to the practicable minimum.

3.6.2 Foolproofness. Where improper installation of an item could cause malfunctioning of that item or the system in which it is installed, an unsymmetrical mounting means shall be provided where practical. This mounting shall be so designed that the item can only be installed in its proper operating position. If an unsymmetrical mounting means is not practical, the item shall be so mounted that its proper operating position can be readily and visibly determined by service maintenance personnel.

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Use MIL-H-5606 Hydraulic Fluid

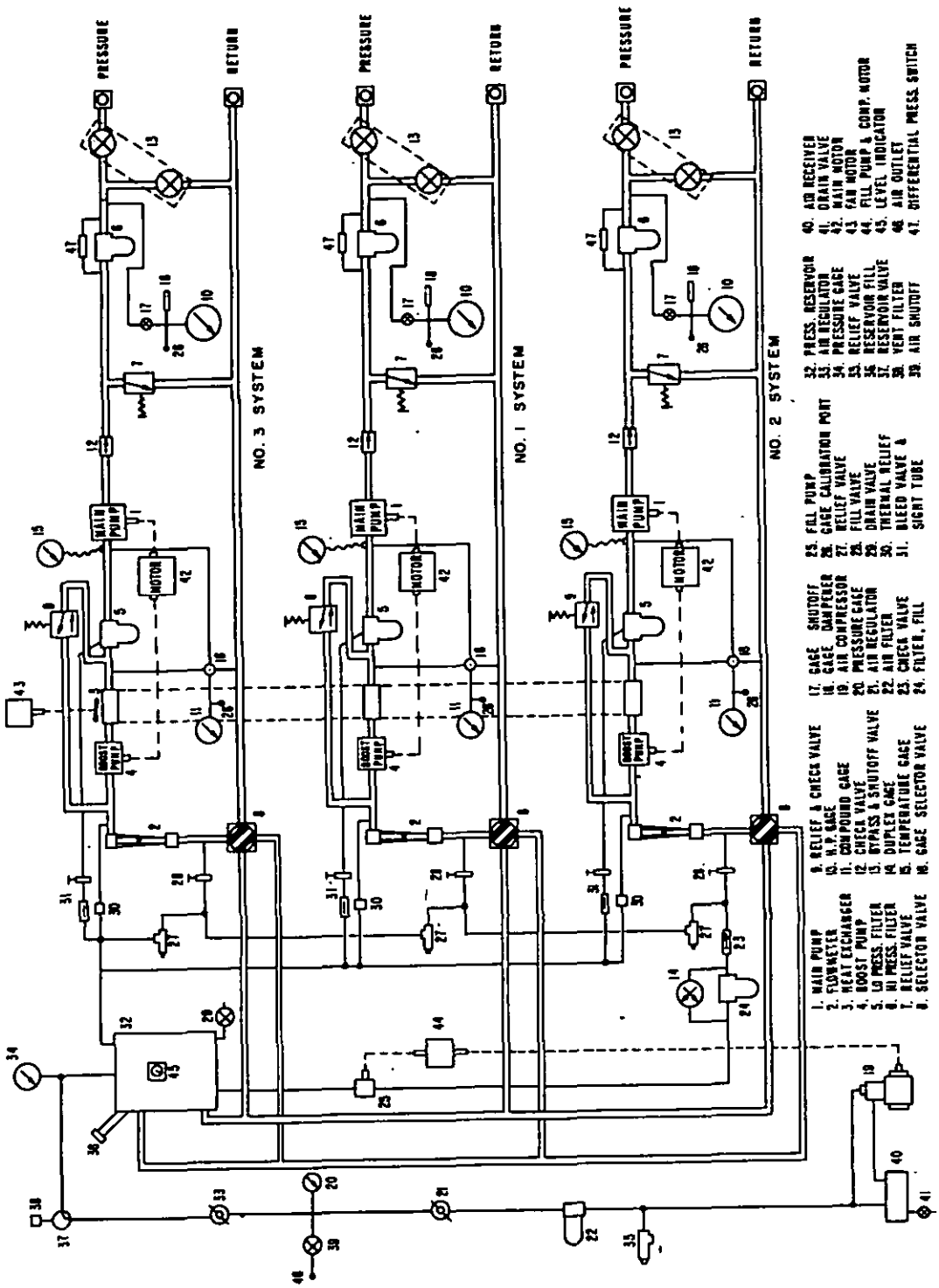


FIGURE 1. Schematic of Hydraulic System

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3.7 Performance. The test stand shall be capable of operating under and satisfactorily resisting the following:

a. Temperatures ranging from -40° to +125°F

b. Fungus growth as encountered in tropical climates

c. Sand and dust particles as encountered in desert areas

d. A relative humidity of 95 to 100 percent at an operating temperature of 125°F and a storage temperature of 160°F

e. Exposure to salt fog

f. Rainfall as encountered in any locale.

3.7.1 Storage temperatures. The test stand shall withstand storage temperatures ranging from -65° to +160°F.

3.7.2 Operating positions. The test stand shall function satisfactorily when its normal operating plane is at an angle of 8-1/2° in any direction from true horizontal.

3.8 Hydraulic systems

* 3.8.1 High-pressure pumps. Three axial-piston-type, high-pressure pumps shall be provided. They shall have aircraft pump impulse characteristics by using nine pistons and high operational speeds (1,500 to 4,000 rpm inclusive). The pumps shall be of the variable-stroke, variable-pressure, pressure-compensated type with an adjustable maximum volume control. They shall be of sufficient rating to meet all requirements specified herein. Each pump shall provide not less than 20 gpm at 3,000 psi and 10 gpm at 5,000 psi at 3,600 rpm.

* 3.8.1.1 The high pressure pumps shall be mounted on the motors by means of a mounting in accordance with AND20006. Each pump shall require not more than 40 hp from the motors specified in 3.9.1 when continuously pumping hydraulic fluid conforming to MIL-H-5606 at 20 gpm and raising the fluid pressure from 110 to 3,000 psia at 68°F.

* 3.8.1.2 The pump cases and components shall withstand and operate throughout inlet-fluid temperatures ranging from -40° to +250°F and pressures from 110 to 265 psi. The pumps shall have a minimum life expectancy of 1,000 hours of continuous duty at full rated load, certified by the pump manufacturer and confirmed by testing.

3.8.1.3 The pumps shall have adjustable pressure-compensator controls capable of regulating the pressure at a constant value with the flow varying from 0 to 20 gpm. The discharge pressure of the pumps shall be 2,700 ±50 psi when delivering 20 gpm and when the zero flow pressure setting is at 3,000 psi. The variable volume controls and pressure controls shall be mounted on the control panel. The speed of compensation of the pumps shall be equal to that of aircraft-type pumps (approximately 0.05 second). The adjustment range of the pressure compensators shall be from 750 to 5,000 psi. The pump shall meet the performance characteristics shown on figure 2 and 3.

3.8.1.4 Fixed relief valves shall be incorporated to maintain the pump case pressure at a value recommended by the pump manufacturer.

* 3.8.1.5 The pumps shall be furnished with flanges and spline drive conforming to AND10266, type XVI-C. The pumps shall turn counterclockwise when viewed from the shaft end. They shall provide a range of 30 inches Hg vacuum to 300 psi.

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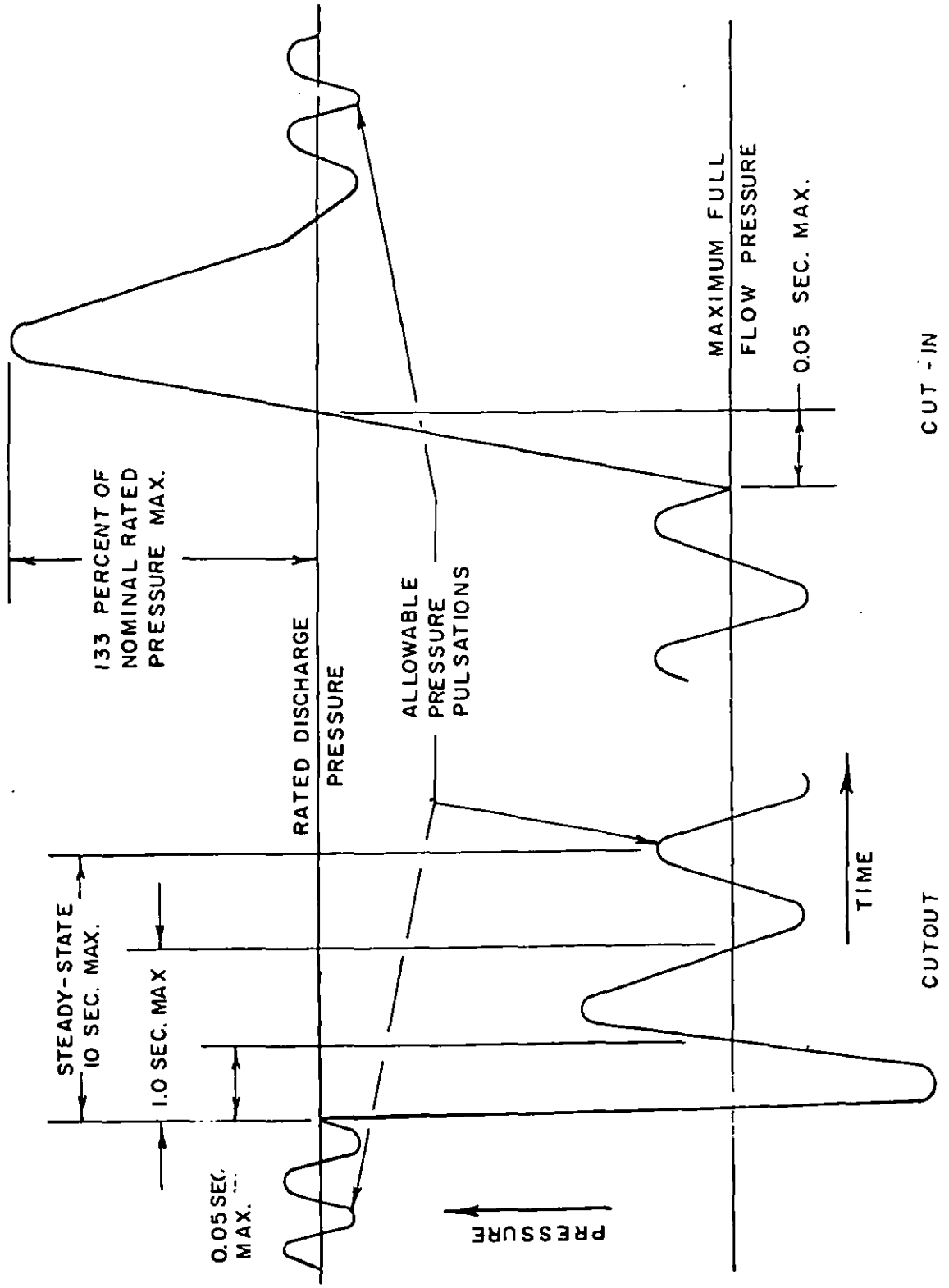


FIGURE 2. Typical Pressure vs Timechart

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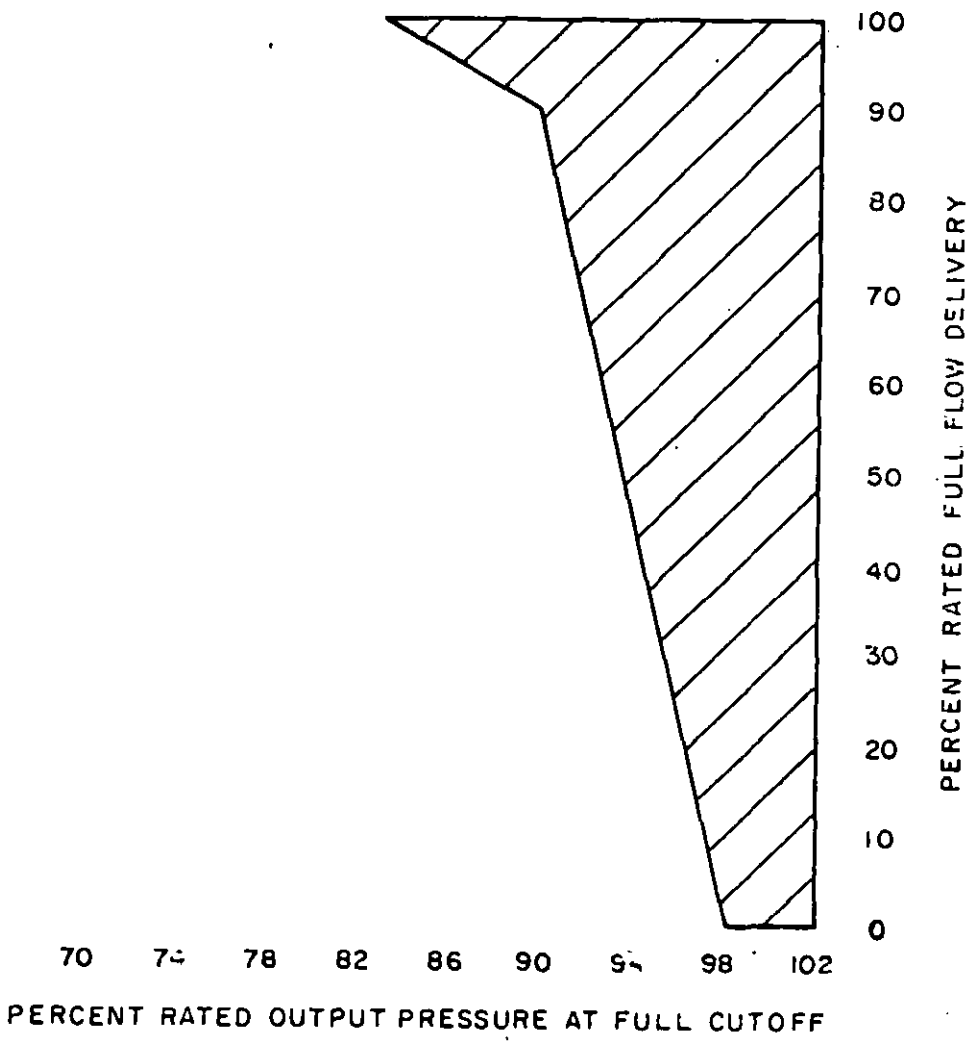


FIGURE 3. Flow Characteristics

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3.8.2 Low-pressure boost pumps. Three low-pressure hydraulic pumps shall be provided to meet the minimum life expectancy requirements of the high-pressure pumps. Each boost pump shall continuously supply at least 25 gpm to its respective high-pressure pumps with inlet pressures at 20 inches Hg absolute and outlet pressures not less than that recommended by the high-pressure pump manufacturer, at all flow and pressure conditions specified herein. Check valves shall be located as shown on figure 1 to permit fluid to bypass the boost pumps. The check valves shall be of the 1-1/2-inch, quick-acting, poppet type with a maximum of 2 psi pressure drop when passing 20 gpm.

* 3.8.2.1 Interlock system. An interlock system shall be incorporated into each hydraulic system that will shutdown the system if the boost pressure falls below the recommended pressures. The interlocks shall prevent damage to the high-pressure pumps that might otherwise result from the loss of boost pressure.

3.8.3 High-pressure filters. Three high-pressure filters capable of operating at 5,000 psi shall be provided. Each filter shall conform to MIL-F-27656, as shown on figure 1.

3.8.3.1 Differential pressure indicators. Backpressure differential pressure indicators shall be connected into the system as shown on figure 1. They shall indicate when the pressure drop across the high-pressure filter is 100 ±15 psi. The indication shall be the closing of a switch that illuminates a red warning light mounted on the instrument panel in a conspicuous location.

3.8.4 Low-pressure filters. Suction line low-pressure filters shall be provided and located as shown on figure 1. Each filter shall use a minimum of one replaceable AN6236-3 filter element or two replaceable AN6236-2 filter elements. Drain plugs shall be provided. Clearance space shall be permitted around each item for draining and replacing the filter elements. The low-pressure filters shall be located on a level above any other part of the line between the boost pump and high-pressure pump so that the entrapped air will collect within the filter and will not be discharged into the high-pressure pump inlet line.

* 3.8.4.1 Filter bleed valves. Pushbutton-type, normally closed bleed valves shall be mounted on the control panel. They shall serve to bleed entrapped air to the test stand reservoir from the top of the low-pressure filter cases or any other locations at which air may be more apt to collect. The valves shall also serve to bleed or drain the aircraft reservoir, when desired. Force necessary to operate these valves shall not exceed 10 pounds. When in the released position, the valves shall not permit leakage from the aircraft hydraulic system into the reservoir to exceed 2 drops in 5 minutes. Valve and tubing size shall be not less than 3/8 inch.

3.8.4.2 Sight tubes. Sight tubes shall be provided in the bleed lines as shown on figure 1. They shall be mounted on the control panel adjacent to the bleed valves. The sight tubes shall withstand a 200-psi working pressure and 300-psi proof pressure. They shall be 3/8 inch tube size, and 3 inches minimum visible length. The sight tubes shall be lighted from behind.

* 3.8.4.3 Thermal pressure relief valves. Preset relief valves shall be provided and located in each hydraulic system as shown on figure 1. The valves shall relieve pressures in excess of 150 ±25 psi and thus preclude damage to components in the low-pressure hydraulic circuits.

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3.8.5 Relief valves. High- and low-pressure relief valves shall be provided and located in the system as shown on figure 1. The relief valves shall be located as near as practical to the high-pressure pump inlet and outlet.

3.8.5.1 High-pressure relief valves. The high-pressure relief valves shall be mounted on the control panel. They shall be adjustable by hand from 500 to 5,000 psig. The high-pressure valves shall incorporate locking devices so that vibrations cannot change the settings.

3.8.5.2 Low-pressure relief valves. The low-pressure relief valves shall be of the balanced type, sensitive to back-pressure. They shall be located as near the high-pressure pump inlets as practical. The low-pressure relief valves shall be adjusted to a fixed pressure of not less than 110 nor more than 265 psi. The low-pressure relief valves shall close to zero flow in 0.025 second and shall have a pressure drop no greater than 25 psi when passing 16 gpm.

3.8.6 Reservoir. A stainless-steel or aluminum hydraulic-fluid reservoir shall be mounted within the test stand housing. The reservoir shall be of sufficient capacity to supply at least 25 gallons of fluid to the systems when inclined 8-1/2° from vertical in any direction. It shall have additional space for a 3-gallon expansion above the normal full level and at least a 2-gallon space below the outlet to the system for collecting moisture and impurities. The reservoir shall be designed for maximum operating pressures of 100 psi and structural proof tests of 200 psi.

3.8.6.1 The reservoir shall conform to MIL-R-5520 regarding obstruction to flow, structural design, shell, seams, structural strength, baffles, fluid connections, sump, drain, return line inlet, scuppers, filler unit, finish, and filling time. Clearance shall be provided around the filler opening to permit rapid filling from 1-gallon cans while the test stand is in operation. The reservoir fill cap shall be either the type that cannot be removed while the reservoir is pressurized or the type that will automatically depressurize while it is being removed.

3.8.6.2 Fluid-level indicator. A dial type fluid-level indicator shall indicate the quantity of available fluid in the reservoir to within ±2 gallons of actual quantity. The indicator shall be mounted on the instrument panel.

3.8.6.3 Selector valves. Four-way, two-position selector valves shall be provided as shown on figure 1 to either route the return fluid to and from the reservoir or to cause the return fluid to bypass the reservoir. These valves shall be shear seal valves and shall be capable of operating in pressures varying between 20 inches Hg vacuum and 200 psi. These valves shall be mounted on the instrument panel.

3.8.6.4 Pressure regulator. A commercial air pressure regulator of the relief type shall be provided and shall be modified for satisfactory operation from -40° to +125°F ambient temperature. The pressure regulator shall control the pressure to the reservoir at any setting from 0 to 100 ±10 psi and maintain the set pressure at any flow including zero flow. The pressure regulator shall be equipped with a variable relief valve adjustable from 0 to 100 psi and shall be installed between the regulator and reservoir.

3.8.6.4.1 Pressure gage. A pressure gage shall be provided to indicate pressure in the reservoir. The pressure gage shall be accurate to within 5 percent of full-scale range.

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3.8.6.5 Pressurization valve. A three-way, two-position valve shall be installed between the pressure regulating valve and relief valve. In one position, the reservoir shall be pressurized; while in the other position, the reservoir shall be exhausted and the pressure regulating outlet blocked off.

3.8.6.6 Drain valve. A drain valve shall be provided for draining the reservoir. Necessary clearance shall be provided around the drain outlet to permit rapid draining into 1-gallon containers.

3.8.6.7 Access plates. One or more access plates of at least 7 inches diameter or 8 by 6 inches in accordance with standard design practices for ASME coded vessels shall be provided to permit visual inspection and manual cleaning of the interior of the reservoir.

3.8.7 Pressure outlets. Three pressure outlets shall be provided as shown on figure 1. Each outlet shall be color-coded to correspond with the color code on the control panel. The outlets shall not be located where leakage of fluid from the outlets could fall on the trailer tires or within the housing. The outlets and dust caps shall conform to Aeroquip part No. 015628-S2-12 and No. 155-S7-12D, respectively, or a part equal to and interchangeable with the Aeroquip part.

*3.8.7.1 Pressure connections. All valves, connections, and tubing used from the high-pressure pump discharge ports to the ends of the 30-foot pressure hoses shall be of a size that will not permit the pressure drop to exceed 150 ±10 psi with 20 gpm flow. All connections and fittings on the high-pressure side shall be not less than 1-inch tubing size. Tubing shall be used on all high-pressure lines within the test stand, except where impractical because of vibration.

*3.8.8 Return connections. Three return connections shall be provided as shown on figure 1. The outlets shall conform to Aeroquip part No. B145-S2-16D or a part equal to and interchangeable with the Aeroquip part. A dust cap conforming to Aeroquip part No. 145-S7-16D or a part equal to and interchangeable with the Aeroquip part shall be attached to the outlet. Each outlet shall be color-coded to correspond with the color code on the control panel. The outlets shall be located for convenience. The locations shall not permit leakage from the connections to fall on the trailer tires nor within the test stand housing.

3.8.8.1 Low-pressure components and connections. The size of all the connections used in the suction side of the hydraulic system shall be such as to allow not more than a 20-psi drop at 20 gpm with fluid temperatures of 60°F from the aircraft outlets through 30 feet of hose to the suction side of the boost pumps. Suction lines and components shall be not less than 1-1/4 inch tubing size and rated no less than 200 psi.

3.8.8.2 External and internal leakage. Total external leakage shall not exceed 2 drops per hour at any flows or pressure specified herein. Internal leakage shall not exceed 5 cubic inches per hour from the closed loop system of either system.

3.8.9 Oil coolers. An air-to-oil cooler(s) shall be provided in the systems as shown on figure 1. The oil cooler(s) shall maintain the oil temperatures of each system below 140°F when 20 gpm at 750 psig is bypassed through the coolers with an ambient air temperature of 100°F. The oil cooler(s) shall permit full flow through the bypass without damage resulting therefrom. The oil cooler(s) shall operate under

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pressures of 200 psi without leakage or damage. The oil cooler(s) shall be provided with a means to make them inoperable in low temperatures.

3.8.10 Flow control and bypass valves. Three flow control and three bypass valves of 1 inch size, or larger, shall control or stop the flow into or from each pressure outlet. The valves shall withstand 10 gpm flow and 5,000 psi pressure. They shall accurately meter the flow throughout the volume range of 0 to 20 gpm and shall be pressure balanced so that a force of 40 pound-inches is the maximum required turning force at 3,000 psi. The valves shall go from the closed position to the open position in not more than two full turns of the handle. The valves shall control the flow to within ±1/4 gpm where flow rates are below 5 gpm, and ±1 gpm with flow rates from 5 to 20 gpm.

* 3.8.11 Fill system. A fill system incorporating a 3-phase, 440-V electric-motor-driven pump, filter, pushbutton-type fill valves, and relief valves, as shown on figure 1, shall be provided to facilitate adding oil or filling each system as desired.

3.8.11.1 Filling pump. The filling pump shall be a low-pressure, hydraulic pump having a capacity of at least 2 gpm at 150 psi. The filling pump and motor controls shall be in accordance with 3.10.3 and 3.10.4.

3.8.11.2 Filling valves. Three pushbutton-type valves shall be mounted on the control panel. Force necessary to operate the valves shall not exceed 10 pounds. When in the release position, the valves shall not permit leakage from the aircraft hydraulic system into the reservoir to exceed 10 drops in 5 minutes. Valve size shall be not less than 3/8 inch tubing size.

* 3.8.11.3 Fill system relief valves. Three variable relief valves, as shown on figure 1, shall be provided (but not mounted) on the control panel. The valves shall be adjustable throughout a pressure range of 0 to 140 psi and capable of relieving the total volume discharged by the fill pump. The valves shall have a locking device so that vibrations cannot change the pressure settings. The relief valves shall be so designed that maximum pressure settings will be 140 ±5 psi (set at 110 psi).

3.8.11.4 Fill system check valve. A check valve shall be located in the fill system as shown on figure 1 to prevent reverse flow to the fill system pump.

* 3.8.11.5 Fill system filter. An AN6235-2A filter element shall be located in the fill system as shown on figure 1. The filter assembly shall conform to an MS28720-6 filter configuration, except it shall be suitable for 145 psi operating pressure.

3.8.11.6 Fill system connections. All lines, tubing, hoses, and connections used in the fill system shall be 3/8 inch tubing size or larger.

3.8.12 Pressure check valves. Pressure check valves that are dimensionally interchangeable with AN6249-16 valves, but suitable for operation at 5,000 psi, shall be located as near the high-pressure pump outlet as practical to prevent reverse flow to the high-pressure pumps. (See figure 1.)

* 3.8.13 Flowmeters. Three flowmeters of the transparent type, rated at 300 psig operating pressure, and having a range of

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3 to 30 gpm and an accuracy of ±2 percent of full scale reading shall be provided in the low-pressure lines as shown on figure 1. A 250-mm minimum length readable section shall be illuminated for continuous visual inspection of the fluid for air. The flowmeters shall be easily removable for cleaning without affecting calibration.

3.8.14 High-pressure gages. Flush-mounted temperature-compensated high-pressure gages shall be provided. They shall conform to GG-G-76, except where conflicts exist the provisions of this specification shall govern.

3.8.14.1 The gages shall be constructed of corrosion-resistant materials. They shall have a 6-inch diameter dial and a steel-alloy bourdon tube. The pointers shall be adjustable without removing the gages from the panel. The gages shall withstand operation at -40° to +350°F.

3.8.14.2 The gages shall provide a range of 0 to 6,000 psi and shall incorporate a means to restore their accuracy under all service conditions. The gages shall be accurate to within 1/2 percent, or less, of full scale at all temperatures and conditions specified herein.

3.8.14.3 The gages shall be of the recess type provided with a mounting flange at the front of the gage. The mounting flange and dial shall not extend more than 5/8 inch beyond the external surface of the panel.

3.8.14.4 The gages shall incorporate a safety blowout device. The safety blowout device shall automatically relieve the gage case pressure should it reach 1/2 psi due to gage failure.

3.8.14.5 Stainless-steel movements shall be provided on all gages. The dial and pointer shall be separated from the gage movements by a wall that will prevent rupture through the front of the gage in the event of gage failure. Gages subjected to continuous surges and extreme pressure fluctuations shall be protected by a hydraulic dampener.

3.8.14.6 Mounting. All gages shall be mounted by screws and tapped panel holes to permit removal by one man from in front of the panel.

3.8.14.7 Connection. All fluid gages shall be connected with flexible AN- or MS-type hose, or commercial equivalent, of sufficient length to permit removal and installation of the gage from the front of the panel.

3.8.14.8 Throttling. All pressure gages shall be equipped with needle valves located as close as practicable to the gages.

3.8.14.9 Gage lines. The high-pressure gage lines shall be provided with needle shutoff valves, accumulator-type pulsation dampeners, and gage test fittings.

3.8.14.9.1 Shutoff valves. The needle shutoff valves shall be of the panel-mounted type with corrosion-resistant steel stems. The valves shall show the manufacturer's cold-working pressure rating stamped on the valve body with an arrow indicating the flow direction. The valves shall withstand working pressures up to 5,000 psi.

3.8.14.9.2 Gage test fittings. The following fittings shall be mounted on the control panel as shown on figure 1. They shall be mounted adjacent to the appropriate gage. The control panel shall be

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marked to indicate that the fittings are gage test connections.

a. An AN832-4D nipple

b. An AN924-4D nut

c. An AN929-4 cap.

* 3.8.15 Low-pressure gages. Flush-mounted compound gages shall be provided and shall have corrosion-resistant-steel bourdon tubes and dials of 3-1/2 inch diameter or larger. The gages shall be accurate to within 1 percent of full scale range and shall conform to GG-G-76, except that where conflicts exist, the provisions of the specification shall govern. They shall provide a range of 30 inches Hg vacuum to 300 psi. The gage movements shall be constructed of stainless steel throughout and shall withstand operation from -40 to +250°F. In all other respects, the gages and mountings shall conform to 3.8.14 through 3.8.14.7 and 3.8.14.9.2.

3.8.15.1 Gage selector valves. Three shutoff-type selector valves shall be mounted on the instrument panel. They shall permit pressure selection at the various points as shown on figure 1.

3.8.16 Temperature gages. Flush-mounted, remote-reading temperature gages shall be provided. The range of the gages shall be approximately from -20° to +250°F and shall indicate within ±5°F of the temperature of the fluid supply to the high-pressure pumps. The gages shall have 3-inch diameter dials, or larger, and shall be flush-mounted. The dial faces shall be marked with a red sector as a warning at 160°F and higher. The accuracy of the gages shall not be affected by continued exposure to the climatic extremes specified herein.

* 3.8.17 Duplex gage. A duplex gage with a 3-1/2-inch dial face shall be mounted on the test stand panel to simultaneously indicate the pressure of both sides of the filter in the fill system. The duplex gage shall have a range from 0 to 200 psi and shall be constructed of corrosion-resistant material throughout.

* 3.8.18 Horn. A horn shall be provided which will sound a warning signal when either pressure or temperature conditions, or both, are beyond recommended ranges. The horn shall be audible at 25 feet from the test stand when the test stand is operating at full capacity.

3.8.18.1 Thermoswitches. Adjustable and snap-action-type preset thermoswitches shall be provided to sound the warning horn when the fluid inlet temperature to any high-pressure pump exceeds 165°F. The thermoswitches shall have an adjustable range of 150° to 200°F.

* 3.8.18.2 Pressure switches. Pressure switches shall be provided to shutoff each system when the output pressure at the booster pumps falls below normal.

3.8.19 Hose assemblies. The hose assemblies specified in table I shall be provided with each test stand and shall be equipped as specified.

3.8.19.1 All dust plugs and caps shall be attached to the test stand or respective hoses by metal chains to prevent loss.

* 3.9 Electrical system. Electrical equipment and wiring shall conform to Pamphlet 70 (see 6.2). Electrical equipment less than 18 inches from the ground shall be explosionproof for class I, division 2 location in compliance with section 5115 of Pamphlet 70. The electrical system shall be designed for operation on 440-V, 3-phase, 60-cycle current. The voltage

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* TABLE I

Hose Assemblies

Hose Assembly Part No.	Quantity	Hose Ends (One or Both)	Equipped with
MIL-H-8788 size 12 hose 30 feet long rated 5,000 psi, equipped at both ends with Joint Industry Committee swivel-action-type fittings	3 ea	Both	*Coupling half, Aeroquip part No. 015828-S5-12 *Dust plug, Aeroquip part No. 155-S9-12D
MS28741-16-3600	3 ea	Both	*Coupling half, Aeroquip part No. 145-S5-16D *Dust plug, Aeroquip part No. 145-S9-16D
*Or a part equal to and interchangeable with the Aeroquip part.			

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tolerance shall be ±10 percent and the frequency ±5 percent. A master switch shall be added to the electric system in the dead cable circuit to provide a means to disconnect the facility electric power to the test stand.

3.9.1 Pump motors. Three design B or C, code D or E, ball-bearing, squirrel-cage induction motors not less than 40 hp continuous rating conforming to NEMA Standard MG-1 shall be provided for driving the high- and low-pressure pumps. The pump motors shall conform to CC-M-641 and shall be of sufficient rating to continuously operate the test stand under any conditions specified in section 4, at maximum capacity without exceeding the NEMA recommended temperature limits for the class of insulating materials used.

* 3.9.1.1 The manufacturer's recommended full-load current shall not be exceeded. The pump motor shall be capable of starting on part winding for 440-V operation.

* 3.9.2 Pump motor control. The test stand shall be provided with a remote-operated, magnetic motor control system in accordance with figure 4, providing motor overload protection and circuit protection. The remote-operated control shall be powered by 120V, obtained by reducing the input voltage with a 480/120V transformer. Voltages greater than 120V shall not be used on the control panel.

* 3.9.2.1 Motor starting. The control circuit shall be such that any one of the three motors can be started independently on 440V connection. The motor starters will provide two-step part-winding start on 440V power. Each of the three 40-hp motors shall have independently controlled circuit stop-start pushbutton stations. Each control circuit shall be equipped with interlocks to protect against simultaneous start of more than one large motor to prevent an excessive inrush amperage. All control apparatus, except the pushbutton on-off control buttons and pressure switches, shall be located in a sheet metal enclosed cabinet.

* 3.9.2.1.1 Overload relays. An overload relay shall be provided on each ungrounded conductor for all motors. These relays shall be enclosed and so located that after opening they can be manually reset without opening the enclosure.

* 3.9.2.2 Pushbuttons. A start-stop, momentary-contact pushbutton switch shall be provided and mounted on the instrument panel of each system. The start buttons shall incorporate cutouts for the boost pump interlock system specified in 3.8.2.1. The pushbuttons shall be located and grouped with the hydraulic controls of the respective system.

3.9.2.3 Circuit breakers. A circuit breaker rated in accordance with Pamphlet 70 shall be provided for each circuit or circuit branch.

3.9.2.4 Wiring. Wiring between electrical components shall be in accordance with Pamphlet 70. Number or color codes shall be used on all wires and terminals.

3.9.2.5 Hourmeter. A counter-type hourmeter having a range from 0 to 9,999 hours shall be provided to record the hours of operation of the pump motors.

* 3.9.2.6 Electric cable and connector. The alternating-current input cable shall be an extra-flexible, 6-conductor cable in accordance with MIL-C-5756. The cable shall be 50 feet in length. The cable shall use three conductors not smaller than AWG No. 000, connected to contacts L_1, L_2, and L_3; one conductor not smaller

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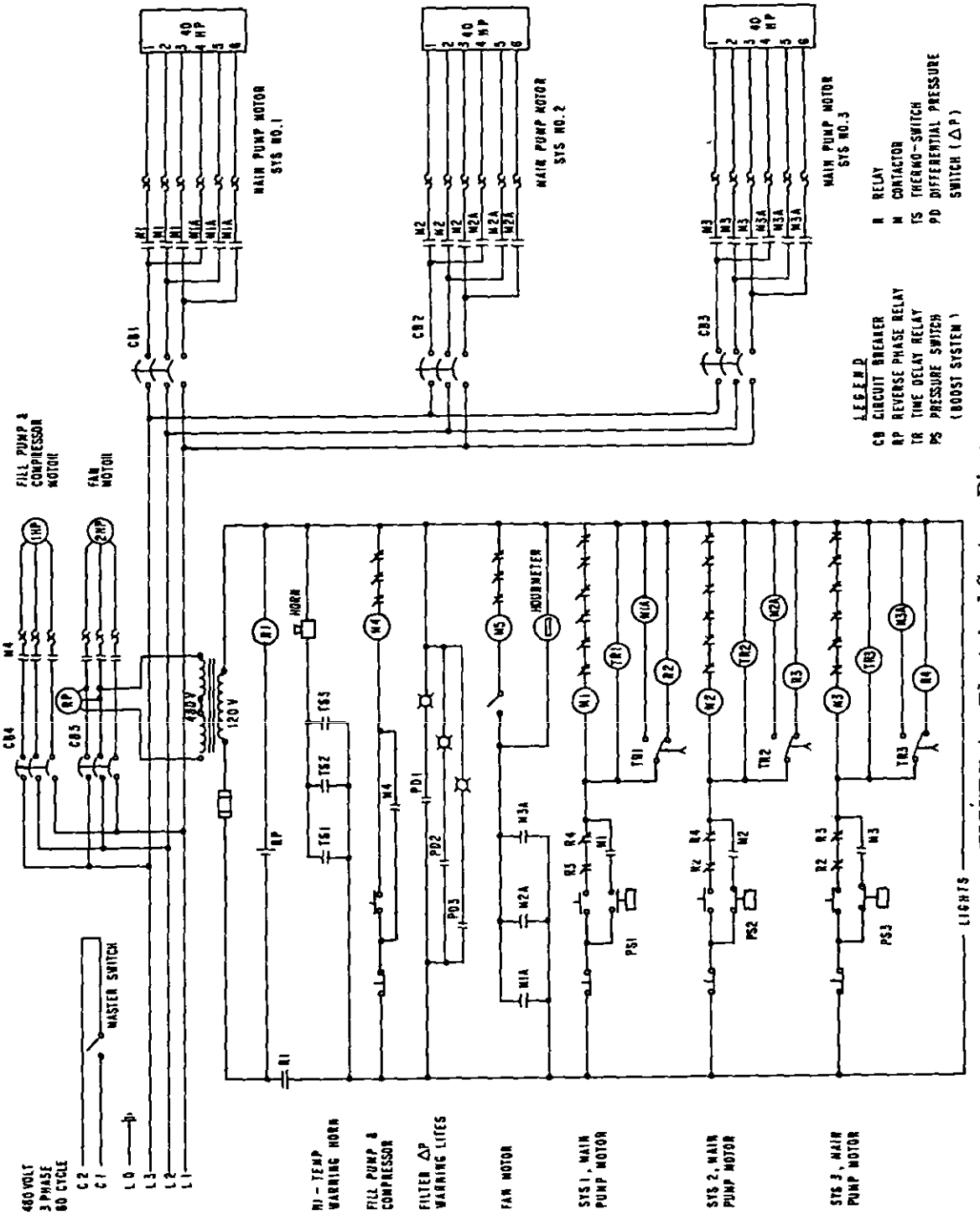


*FIGURE 4. Electrical System Diagram

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than No. 8 connected to contact L_0 (ground); and two conductors not smaller than No. 12 connected to contacts C_1 and C_2. The connector shall be a watertight, molded-rubber, polarized type conforming to MS27265. The cable shall be securely attached to the test stand to remove all strain from the terminals. Provisions shall be made to secure the cable for storage above the towbar with cable rack.

3.9.2.7 Reverse rotation protection. A mechanical or electrical device shall be provided that will prevent operation of the pump motors in the opposite direction of rotation from that required for the pumps, in the event of incorrect phasing.

* 3.10 Pneumatic pressurizing system. A compressed air supply shall be provided for pressurizing the aircraft or test stand reservoir.

* 3.10.1 Air compressor. A commercial air compressor shall be furnished and shall be capable of continuously providing 3 scfm at 150 psi when supplied with air at 77°F and 29.92 inches Hg absolute. An AN832-4D nipple, AN924-4D nut, and AN929-4 cap shall be provided for the outlet connection. An outlet shutoff valve shall be provided to cut off air to the outlet connection. An air receiver and pilot-operated unloader valve shall be provided in the pneumatic system as shown on figure 1.

* 3.10.2 Regulator. A commercial self-relief-type regulator modified for satisfactory operation at -40° to +125°F ambient temperatures shall be provided. The regulator shall control and limit the pressure to the outlet at any setting from 5 to 150 psi and maintain the set pressure at any flow, including zero flow. A pressure gage having a range from 0 to 200 psi shall be included to indicate pressure in the aircraft reservoir and shall be accurate within 3 percent of full scale range. An air line filter or water separator shall be added downstream of the compressor. An automatic means of draining the water or an adequate capacity to store water shall be included.

* 3.10.3 Compressor motor. A 440-V, 3-phase, 60-cycle motor shall be provided and shall be of sufficient rating to continuously operate the compressor at maximum rated pressure and output without exceeding safe temperature limits for the class of insulation used. The compressor motor shall be connected as shown on figure 4.

* 3.10.4 Compressor motor controller. A manual (or magnetic with pushbutton), 440-V, 3-phase, 60-cycle controller with three overload relays shall be provided for starting, stopping, and protecting the compressor motor. The compressor control shall be mounted on the control panel in an instrument cluster of all pneumatic controls and gages.

3.10.5 A system, including hydraulic motor and control valve, may be used in place of the electric motor and controller specified in 3.10.3 and 3.10.4.

3.10.6 Hose. Thirty feet of AN6270-4 hose in accordance with MIL-H-5593 shall be provided.

* 3.11 Fan(s) or blower(s). A fan(s) or blower(s) with a drive mechanism from the pump motors or driven by a separate electric motor shall be furnished. The fan(s) shall provide sufficient cooling air to the air-to-oil coolers to meet all oil cooling requirements. The fan(s) or cooler(s) shall normally function whenever the pumps are operating, however, for low-temperature operation, provisions shall be made to disconnect or shutdown the drive mechanism for the fan(s) or blower(s).

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3.11.1 The controls or switches shall not be mounted on the control panel, but instructions as to their location and operation shall be provided as part of the directions for low-temperature operation as specified in 3.21.4.

3.11.2 Should an electric motor or drive be used, the requirements specified in 3.9, 3.10.3, and 3.10.4 shall govern motor and control design.

3.12 Trailer. The components of the test stand shall be assembled and mounted on a trailer. The test stand shall meet the design and performance requirements of MIL-M-8090 and those specified for type II, group C mobility.

3.12.1 Frame. The trailer frame shall be of welded-steel construction. The frame shall be not less than 54 inches nor more than 66 inches wide and shall be not less than 84 inches nor more than 108 inches long. The ratio of width divided by length shall not exceed three-fourths.

*3.12.1.1 Static discharge reel. An RMU-18/E static discharge grounding reel assembly conforming to MIL-R-38489 with a 50-foot cable shall be installed.

3.12.2 Springs. The trailer springs shall provide good riding qualities, thereby precluding damage to the mounted equipment. The springs shall be of such construction that not more than 3 inches is added to the frame height for spring deflection.

3.12.3 Parking brakes. Brakes for both rear wheels shall be interconnected and shall operate from one handle.

3.12.4 Lifting and tiedown fittings. Lifting and tiedown fittings conforming to Drawing 48B7796 or 49B6564 shall be provided and shall withstand the loadings specified in MIL-A-8421. The test stand shall be designed for forklifting.

3.12.5 Lubrication. All moving parts shall be provided with suitable means of lubrication and shall operate satisfactorily when lubricated with standard military lubricants as specified in MIL-M-8090.

3.12.5.1 Lubrication fittings. Lubrication fittings shall conform to MIL-F-3541 and shall be located in accessible, protected positions. Extended fittings shall be provided to lubricate parts or assemblies that are not readily accessible for direct lubrication or which are likely to be overlooked because of inaccessibility. The test stand and all components shall be completely lubricated prior to delivery.

3.12.5.2 Lubrication chart. A lubrication chart shall be provided directing attention to all lubrication fittings and shall specify the range and grade of lubricant required for critical temperatures. The chart shall be permanently attached in a protected conspicuous location.

3.12.6 Pintle hook. The trailer shall be equipped with a pintle hook conforming to MS51335 mounted on the center of the rear frame.

3.13 Control panel. A control panel shall be provided for mounting all the meters, valves, indicators, gages, switches, and controls furnished with the test stand. Instruments subject to error from vibration shall be shock mounted. The control panel shall be constructed of No. 12 U.S. gage (0.1046 inch), or thicker, sheet steel, and supported by a welded-steel, self-supporting frame work. The control panel frame shall be bolted or welded to the trailer bed and so positioned that the trailer wheels, tow bars, et cetera, will not interfere with the operator.

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3.13.1 The control panel shall be located at a sufficient height above the ground and inclined at a sufficient angle, 10° minimum, to facilitate convenient observation and operation of all items by an operator approximately 5 feet 10 inches tall.

3.13.2 All controls and gages for the No. 1 system shall be grouped in one section of the control panel and valve packing nuts, handles, buttons, and gage face rings shall be painted red. All controls and gages for the No. 2 system shall be grouped in one section of the control panel and valve packing nuts, handles, buttons, and gage face rings shall be painted green. All controls and gages for the No. 3 system shall be grouped in one section of the control panel and valve packing nuts, handles, buttons, and gage rings shall be painted blue. In lieu of color coding each nut, handle, button, and gage face ring to distinguish systems, the panel may be color-coded.

3.14 Housing. The test stand components and control panel shall be completely enclosed and protected by a weather-resistant housing. The housing shall consist of a welded-steel framework covered with No. 16 U.S. gage (0.0598 inch), or heavier, sheet steel. The housing shall be braced to prevent distortion or excessive vibration during handling and operation of the test stand or handling of the housing while removed from the test stand.

3.14.1 The housing shall support a 200-pound weight distributed evenly over an area 4-1/2 by 5 inches and applied in any location on top of the test stand.

3.14.2 The housing shall be so designed that water or hydraulic fluid spilled any place therein will drain to the exterior and will not collect within the enclosure.

3.14.3 Metal hooks. Metal hooks shall be provided on the exterior of the housing to permit temporary storage of the pressure hoses. The hooks shall be designed to permit flat positioning against the housing when not in use.

3.14.4 Access and ventilating doors. The housing shall be provided with sufficient access and ventilating doors or adjustable louvers to permit access for operating and servicing and to ventilate the test stand. The doors and louvers in the housing shall be free from binding and shall be easy to operate with one hand. Provisions shall be incorporated to rigidly secure the doors in the full-open or full-closed position.

* 3.14.5 The housing shall be easily removable to facilitate ease of maintenance and accessibility.

3.14.6 Compartments. A handbook compartment of 2 by 9 by 12 inches or larger, and an enclosed accessory box of 8 by 8 by 24 inches or larger, shall be permanently attached within the housing. The compartment shall be constructed on the trailer bed or shall be fastened to the housing assembly. Provisions shall be incorporated within the outline of the housing for storage of the hose assemblies and accessories specified herein.

3.15 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

3.16 Electromagnetic interference. The test stand shall meet the requirements of MIL-STD-826, method 1001, class Gp.

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3.17 Noise level. The noise level of the test stand shall not exceed the values in the frequency ranges specified in 4.6.15.

3.18 Dimensions. The length and width of the test stand and housing shall be not less than the trailer frame assembly nor extend beyond the trailer wheels at any point. The overall height of the test stand shall be held at a minimum consistent with good design practices, but shall not exceed 44 inches above the top of the tires.

3.19 Weight. The weight of the test stand, including the trailer, shall not exceed 5,000 pounds.

3.20 Finishes and protective coatings. Cleaning, painting, and plating shall be in accordance with MIL-STD-808.

3.20.1 Tubing and components that have been treated with corrosion-resistant coatings need not be painted.

3.20.2 Color. Color shall be used in accordance with MIL-STD-808; exterior yellow, code DG; interior gray, code LG.

3.21 Operational markings

3.21.1 A caution plate containing the following information shall be securely attached to the instrument panel adjacent to the selector valves:

DO NOT OPERATE WHEN THE FLOW CONTROL VALVE IS OPEN.

3.21.2 An 8- by 10-inch engraved, permanent-type diagram plate, arranged similarly to figure 1 to show the hydraulic circuit of the test stand, shall be securely fastened to the center of the instrument panel.

3.21.3 All items mounted on the control panel shall be identified as to use and cautions by permanently etched or engraved metal or plastic plates securely attached to the panel.

3.21.4 A permanently engraved plate containing low-temperature-adjustment directions for the oil cooler(s) shall be provided on the control panel.

3.21.5 An etched or engraved permanent-type metal plate containing the following information shall be securely attached adjacent to each pressure outlet:

PRESSURE OUTLET NO._____.

3.21.6 An etched or engraved permanent-type metal plate containing the following information shall be securely attached adjacent to each suction connection outlet:

RETURN OUTLET NO._____.

3.21.7 An etched or engraved permanent-type temperature correction chart shall be installed near the flowmeters.

3.21.8 A permanent-type nameplate containing complete instructions for converting the test stand from operation on 240 to 480V or vice versa shall be fastened inside the starting box.

3.21.9 The parking brake control device shall be appropriately marked with operating instructions for release and application of the brakes. The marking shall be in 1/4-inch black letters and shall be visible to the operator.

3.21.10 The exterior of the test stand shall be stenciled to indicate the tire pressure, center of gravity, towing speed, and any other markings necessary for proper handling and maintenance.

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3.22 Identification of product. The following additional markings shall be included:

Rating: A/M27T-2

0 to 20 gpm to 3,000 psi, each system, 0 to 60 gpm total

0 to 10 gpm - 3,000 to 5,000 psi, each system, 0 to 30 gpm total

Towing speed: 20 mph maximum.

3.22.1 Special markings

3.22.1.1 The following information shall be stenciled in 1-inch black letters on each of the four sides of the test stand:

FOR AIR SHIPMENT THIS TEST STAND IS DESIGNED FOR 8G FORE AND AFT, 3G SIDEWARD.

3.22.1.2 Drawbar or housing marking. The towbar and the front of the housing shall be marked in 1-inch black letters with the following notation:

MAX SPEED 20 MPH.

3.23 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.24 Workmanship. The test stand, including all parts and accessories, shall be fabricated and finished in a workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs, and sharp edges; accuracy of dimensions, radii of fillets, and marking of parts and assemblies; thoroughness of soldering, welding, brazing, painting, wiring, and riveting; alignment of parts and tightness of assembly screws, bolts, et cetera.

3.24.1 Riveting. Riveting operations shall be performed to insure tightness and satisfactory heading of rivets.

3.24.2 Cleaning. The test stand shall be thoroughly cleaned. Loose, spattered, or excess solder; metal chips; and other foreign materials shall be removed during and after final assembly.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the test stand shall be classified as follows:

a. Preproduction testing See 4.4

b. Acceptance tests See 4.5.

4.3 Test conditions

4.3.1 Standard atmospheric conditions. Unless otherwise specified herein, tests shall be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately 70°F).

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When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified condition.

4.3.2 Test time. Unless otherwise specified herein, any test period shorter than 1 hour shall not be credited as test time.

4.3.3 Apparatus and instruments. Apparatus and instruments used in conjunction with the testing specified herein shall be of laboratory precision-type and shall be checked to ascertain that they have been installed in a satisfactory manner for durability and accurate operation. They shall be calibrated at intervals properly spaced to insure laboratory efficiency.

4.3.4 Fluid flow. The calibration and performance checks of the displaced fluid shall be determined by means of a calibrated flowmeter with certified accuracy of 1 percent or better of reading.

4.3.5 Preliminary run-in. Preliminary run-in of the test stand as may be recommended by the manufacturer shall be performed by the manufacturer prior to submission of the test stand for testing. A certified log of all preliminary running time, calibration data, et cetera, shall be submitted to the procuring activity.

4.4 Preproduction testing (see 6.2)

4.4.1 Test sample. One test stand shall be subjected to the preproduction tests specified in 4.4.3.

*4.4.2 Test report. Upon completion of the preproduction tests, a test report shall be prepared in accordance with MIL-STD-831 and three complete copies of the report furnished the procuring activity.

*4.4.2.1 Reliability and maintainability information. The following information shall be included as an appendix to the test report:

a. All failures, servicing, adjustments, maintenance, and irregular functioning shall be identified by accumulated operating time, cycles, miles, or position in the test procedure, as appropriate. Test conditions at the time of the events identified shall be recorded

b. A summary of the engineering analysis and of any tests conducted to determine assignable causes for any failure or irregular functioning

c. A summary of the engineering analysis leading to any corrections made to design, construction, quality control, or other procedures, or leading to any corrections to be made or proposed to be made to production items. The summary shall also include an analysis of the predicted effectiveness of such corrections

d. Clock time and man-hours required for each maintenance and servicing action taken during the tests. A brief description of the experience and qualifications of the personnel taking such actions

e. Test activity or contractor comments on item features or requirements that, if modified, should improve the item

f. Test activity or contractor comments on use or maintenance conditions to be avoided or cultivated to increase the reliability or useful life of the item

g. Any of the above information that is already included in the test report body need not be repeated in the information required by this paragraph, but clear reference to the location of the data shall be included.

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4.4.3 Preproduction tests. Preproduction tests shall consist of all tests described under 4.6, except 4.6.23 and 4.6.24.

4.5 Acceptance tests. The acceptance tests shall consist of the following:

a.	Individual tests	See 4.5.1
b.	Sampling plan and tests	See 4.5.2

4.5.1 Individual tests. Each test stand shall be subjected to the following tests as described under 4.6:

a.	Examination of product	See 4.6.1
b.	Calibration test	See 4.6.2
c.	Flow tests	See 4.6.3
d.	Loading pressures	See 4.6.4
e.	Fluid temperature test	See 4.6.5
f.	Road test	See 4.6.23.

4.5.2 Sampling plan and tests

4.5.2.1 Lot. The lot definition, formation, and size shall be in accordance with MIL-STD-105.

4.5.2.1.1 MIL-STD-105. When MIL-STD-105 specifies an action by the Government it shall, at the option of the Government, be performed either by the Government or by the contractor under the supervision of the Government inspector.

4.5.2.2 Tests. Samples selected shall be subjected to the following tests:

a.	Semi-endurance test	See 4.6.6
b.	Examination	See 4.6.7
c.	Maneuverability test	See 4.6.24.

4.6 Test methods

4.6.1 Examination of product. The test stand shall be inspected to determine compliance with requirements specified herein with respect to parts, accessibility, ease of maintenance, materials, workmanship, and markings.

* 4.6.2 Calibration test. The pressure drop from the outlet of the high-pressure pump to the end of the 30-foot pressure hose and the pressure drop from the end of the 30-foot return hose to the suction side of the boost pump shall be measured and shall not exceed 150 psi on the pressure side or 20 psi on the return side.

4.6.3 Flow tests. The constant-flow operation of each system shall be checked at 800, 1,500, and 3,000 psi at each rate of flow of 4, 8, 12, 16, and 20 gpm. The pressure gages shall be accurate to within 1 percent of full scale at all settings. The flowmeters shall be accurate to within ±2 percent of actual flow. The test stand shall then be operated at 30 gpm and 5,000 psi for 1 hour and inspected for evidence of leakage of fluid.

4.6.4 Loading pressures. The operation of each system shall be checked at 1,500 and 3,000 psi at each rate of flow of 10 and 20 gpm and compensation set at 1,500 and 3,000 psi respectively. The hydraulic pumps shall put out full 10 and 20 gpm flow at all loading pressures below the 1,500 and 3,000 psi respectively, but flow shall fall to approximately zero as soon as the loading pressure reaches the compensation pressure setting.

4.6.5 Fluid temperature test. The temperature of the fluid entering each high-pressure pump on the test stand shall be raised above 160°F and the thermoswitch shall sound the warning horn before the

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fluid temperature exceeds 165°F but not before it reaches 160°F. With the test stand operating, the warning horn shall be audible 25 feet from the test stand in any direction.

4.6.6 Semi-endurance test. The test stand shall be subjected to 20 percent of the endurance test specified in 4.6.10. During the 20 hours operation at 20 gpm and 3,000 psi with suction pressure to the low-pressure pump reduced to 20 inches Hg absolute (see 4.6.10c), the boost pressure shall be lowered below the recommended pressure and the operation of the interlock system shall be checked. The interlock system shall shut down the test stand without damaging the high-pressure pump or other parts of the test stand.

4.6.7 Examination. At the completion of the test specified in 4.6.6, the test stand shall pass the tests specified in 4.6.3 and 4.6.4.

4.6.8 Initial disassembly inspection. The test stand shall be disassembled to the maximum practical extent and inspected, the breakdown weights determined, and measurements made that may be used to determine wear. All gages, meters, and instruments shall be laboratory checked against certified calibrating equipment and failure of any item to meet the accuracy requirements specified herein, shall be cause for rejection. All requirements that may be determined by weight or dimensional measurements and inspection shall be checked for compliance, and failure to meet any of these requirements shall be cause for rejection. Following this inspection, the test stand shall be reassembled and subjected to the remainder of the tests.

4.6.9 Environmental tests. The test stand shall be subjected to the following environmental tests conducted in accordance with the specified procedures of MIL-STD-810. Unless otherwise specified herein, upon completion of each test, the test stand shall be subjected to the tests specified in 4.6.3 and 4.6.4. Failure of the test stand to operate as required during or after each exposure, failure of any item or assembly due to deterioration from exposures in any test, or formation of any undesirable condition that cannot be remedied by routine maintenance shall be cause for rejection.

* 4.6.9.1 Low temperature. The test stand shall be subjected to low temperature in accordance with method 502, procedure I, except the storage temperature shall be -65°F and the test stand shall not be operated. The test circuit shall include three MS28759-12-3600 pressure hoses, three restrictor valves to buildup 3,000-psi pressure, one reservoir filled with not less than 40 gallons of hydraulic fluid conforming to MIL-H-5606, and three MS28741-16-3600 return lines. The entire test circuit shall be contained within the cold chamber. Prior to starting the test, the temperature shall be raised and stabilized at -40°F. The fluid in the reservoir shall not be heated from any external source prior to starting or during the test. After starting, the flow and pressure in each system shall be slowly increased to the rated output of 20 gpm and 3,000 psi. The test stand shall attain rated output within 30 minutes after starting and shall be operated at rated output for not less than 30 minutes. All controls, gages, meters, valves, systems, and accessories shall be checked for satisfactory operation while operating in the -40°F ambient temperature. The high-pressure gage shall be accurate to within ±1 percent of full-scale range.

4.6.9.2 High temperature. The test stand shall be subjected to high temperature in

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accordance with method 501, procedure I. The test stand shall be placed within the high-temperature chamber with the selector valve at the reservoir bypass position, and the internal temperature of the chamber raised to 160°F with an internal relative humidity of not more than 5 percent. As the temperature of the chamber is raised, the thermal pressure relief valves shall maintain the internal pressure of the systems at 275 psi or less. Upon completion of the exposure to high temperature, the ambient temperature shall be lowered to 125°F. The constant flow operation of each system shall be checked at 800, 1,500, and 3,000 psi at each rate of 4, 8, 12, and 20 gpm. The pressure gages shall be accurate to within ±1 percent of full scale at all settings. The flowmeters shall be accurate to within ±2 percent of actual flow. The temperature shall then be reduced to 100°F. The test stand shall be operated at 18 gpm at 750 psi through the bypass circuit for 1 hour. The test stand cooler shall maintain the outlet fluid temperature below 140°F. After this test, the test stand shall be inspected for leakage of fluid.

4.6.9.3 Fungus resistance. The test stand shall be subjected to fungus in accordance with method 508, procedure I.

4.6.9.4 Sand and dust. The test stand shall be subjected to sand and dust in accordance with method 510, procedure I.

4.6.9.5 Humidity. The test stand shall be subjected to humidity in accordance with method 507, procedure I.

4.6.9.6 Salt fog. The test stand shall be subjected to salt fog in accordance with method 509, procedure I.

4.6.9.7 Rain. The test stand shall be subjected to rain in accordance with method 506, procedure I. The test stand shall be positioned on the test chamber floor in its normal operating position. After exposure to rain for 1 hour and while still in the rain test, the test stand shall be started and operated at 20 gpm and 3,000 psi from each system for 30 minutes. Failure of the test stand to start or operate satisfactorily during the rain test shall be cause for rejection. Upon completion of the rain test, all surfaces that might collect moisture shall be inspected to determine adequacy of drainage facilities and any collection of water in excess of 1/8 inch deep or that could be detrimental to the performance of the test stand shall be cause for rejection.

4.6.10 Endurance test. The test stand shall be connected in series to a test rig through 30 feet of pressure hoses and 30 feet of return hoses specified in table I. The test rig shall include a reservoir, restrictors to create test pressures in the high-pressure lines, a restrictor to create the specified vacuum in the return lines, and a cooler(s) to maintain fluid temperatures at approximately 130°F. The ambient air temperature shall be maintained at 75°F, or higher, during the endurance test. Readings of all gages, meters, and instruments on the test stand shall be recorded every hour. Output flow, fluid temperature, fluid pressure, inlet fluid temperature and pressure, motor voltage, amperage, power factor, and kilowatt shall be recorded each hour (see 4.6.12). Endurance test runs shorter than 1 hour shall not be credited to the endurance test. The test stand shall first complete the test specified in 4.6.3 and shall then be subjected to the following 350 hours of endurance testing:

a. One hundred hours operation at 20 gpm and 3,000 psi from each system. Pressure shall be discharged through the outlets and

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returned through the suction lines at pressures of 125 psig (reservoir bypassed). The fill motor and pump shall be operated continuously at 150 psi during this period

b. Fifty hours operation at 10 gpm and 5,000 psi from each system. Pressure shall be discharged through the outlets and returned with suction pressures of 0 to 10 psig

c. One hundred hours operation at 20 gpm and 3,000 psi from each system with suction pressure to the low-pressure pump reduced to 20 inches Hg absolute. Pressure shall be discharged through the outlets and returned through the inlets

d. Fifty hours operation at 10 gpm and 3,000 psi from each system with suction pressures of 10 inches Hg to 0 psig

e. Fifty hours operation at 10 gpm and 1,000 psi from each system with suction pressures of 125 psig.

4.6.10.1 During the endurance test, the flow shall be instantaneously varied from the preset flow to zero flow and immediately returned again to the preset flow. This cycle shall be accomplished one thousand times during each 50 hours of testing. Damage to the pump, pump compensator, flow indicator, gearing, or other components shall be cause for rejection.

4.6.10.2 The test specified in 4.6.3 shall be repeated. This data shall be compared with data recorded in 4.6.10. Results shall indicate that no failure or undue wear to the test stand has resulted from the endurance test.

4.6.11 Electrical equipment tests. The electrical system shall be inspected to determine compliance with 3.9 and all items shall be shown to have Underwriters' Laboratories approval where applicable. During the last hour of the full capacity run specified in 4.6.10(a), the temperature rise in all motor windings shall be determined by the resistance method and if the temperature rise exceeds the safe temperature limits for the class insulation used, as specified by Standard C-50, it shall be cause for rejection. The motors shall be capable of safe temperature rise at 6,000 feet pressure altitude and full load and 104°F.

4.6.11.1 Operation of the thermoswitches specified in 3.8.18.1 shall be checked at least twice and shall sound the warning horn when the fluid to the high-pressure pumps reaches not less than 160°F or not more than 165°F. The warning horn shall be audible at 25 feet from the unit. The thermoswitches shall be made inoperative, and the test stand shall operate satisfactorily for 1 hour under the conditions specified in 3.8.1.2.

4.6.11.2 The pump motors shall be overloaded to 125 percent of full-load current and sustained until the overload relays trip and shut off the motors. The overload relays shall trip before damage is caused to the motor and shall not take longer than 15 minutes to open the main contactor. The same test shall be run on the compressor motor (if electric motor is furnished). Ambient temperature for the motor and control shall be 104° ±9°F.

4.6.12 Air compressor test. The air compressor shall be operated at least 25 percent of the time during the endurance test specified in 4.6.10. The performance of the compressor shall be checked periodically during each preproduction test. Failure of a component part of the pressurizing system shall be cause for rejection. The output of the compressor shall be determined and shall equal or exceed the requirements specified in 3.10.1.

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4.6.13 Pump tests. The following tests shall be performed on one new high-pressure and one new low-pressure pump to establish the 1,000-hour life expectancy:

4.6.13.1 High-pressure pump efficiency tests. The high-pressure pump shall be connected to a calibrated power source and operated at 3,400 rpm to deliver 20 gpm flow at 3,000 psi while pumping hydraulic fluid conforming to MIL-H-5606 at 68°F with inlet pressure of 40 psig. Not more than 40 hp input shall be required to drive the pump.

* 4.6.13.2 High-pressure pump endurance. The high-pressure pump shall be driven at 3,400 rpm for a minimum of 1,000 hours under the following conditions. Fluid temperatures during the 1,000 hours shall be maintained at 80°F 25 percent of the time, 120°F 50 percent of the time, and 160°F the remaining 25 percent of the time for each flow condition. Inlet pressure shall be maintained at the minimum recommended by the pump manufacturer for 10 percent of the time, normal recommended pressure 80 percent of the time, and 300 psig 10 percent of the time for each flow condition. Output flow condition shall be as follows:

Length of Run (Hours)	Output Pressure (PSIG)	Output Flow (GPM)
400	3,000	20
200	5,000	10
200	3,000	8
50	950	14
50	1,500	4
100	3,000	20

By use of a quick-acting (0.1 second) solenoid valve in the pressure line, the pump shall be compensated from the present flow to zero flow and vice versa once each minute during this test. The pump case drain flow reading shall be recorded.

4.6.13.3 High-pressure pump compensation time. Inlet pressure, outlet pressure, and delivery of the pump shall be recorded at 10 gpm and 5,000 psi, 20 gpm at 3,000 psi, and 20 gpm at 1,000 psi to show compensating characteristics of the pump. The pump shall compensate from zero flow to full flow and vice versa in 0.05 second, and the pressure shall not exceed 133 percent of the selected pressure at any time.

* 4.6.13.4 Low-pressure pump tests. The low-pressure pump shall be tested at rated flow and speed for 1,000 hours under the following conditions:

Running Time (Hours)	Inlet Condition	Outlet Pressure (PSIG)
300	125 psig	250
200	30 psig	155
300	20 inches Hg abs	95
100	10 inches Hg to 0 psig	110
100	125 psig	250.

4.6.13.5 Fill pump tests. The fill pump shall be operated and shall deliver 2 gpm against a pressure of 150 psi.

4.6.14 Electromagnetic interference. The test stand shall be tested as a complete unit for radiated interference in accordance with MIL-STD-826, method 1001, class Gp.

4.6.15 Noise level test. The noise level of the test stand shall be measured at head level (5 feet 8 inches) in a circle of 25 feet radius centered on the geometric center of the unit at 12 positions equally spaced every 30° and at head level at the

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approximate operator position while the test stand is operating at 60 gpm and 2,700 psi. The test stand shall not exceed the noise levels shown below. Measurements shall be taken out-of-doors on a paved surface.

Cycles Per Second	Not Exceeding (DB) Operator Position	At 25-Foot Radius
27 to 75	106	91
75 to 150	96	81
150 to 300	88	73
300 to 600	85	70
600 to 1200	84	69
1,200 to 2,400	83	68
2,400 to 4,800	82	67
4,800 to 9,600	81	66.

4.6.16 Valve tests. The flow control valves shall be tested by connecting the test stand outlets to external flowmeters. With the pumps set to operate at 3,000 psi and 20 gpm, the flow control valves shall be operated to determine that they can hold the outlet flows steady at any desired value from 0 to 20 gpm. Failure of the valves to regulate the flows to 1/4 gpm at any setting from 0 to 5 gpm and to 1 gpm at any setting from 5 to 20 gpm shall be cause for rejection. The flow control valves shall be tested for leakage with the pumps operating at 3,000 psi. The force required to turn the valves under the above conditions shall not exceed 40 pound-inches. During these tests, the flow control valve settings shall not be influenced by vibration from the test stand or motors. Leakage shall be cause for rejection.

4.6.17 Test stand operation (tilted position). The test stand shall rest on a platform and shall be kept from rolling by the parking brakes. The platform supporting the test stand shall be tilted in eight different directions, 45° apart, on an 8-1/2° incline with the true horizontal plane. The test stand shall be started and operated at 20 gpm and 3,000 psi from each system for 30 minutes in each position. The test stand shall perform satisfactorily in every respect, and the parking brakes shall prevent the test stand from moving on the platform. While tilted in each of the eight positions, 25 gallons of fluid shall be pumped from the reservoir and then replaced to check the capacity specified in 3.8.6

4.6.18 Support test. A 200-pound weight shall be applied slowly and uniformly to an area 4-1/2 by 5 inches in at least six random positions on top of the test stand housing, including a reading in the center of each panel. The test stand shall not be permanently deformed by the application of the weight.

4.6.19 Mobility tests. The test stand shall be subjected to the mobility tests of MIL-M-8090 as specified for type II, group C equipment. Upon completion of these tests, the test stand shall be subjected to the tests specified in 4.6.3 and 4.6.4. Calibration of gages and minor adjustments will be permitted.

4.6.19.1 Parking brakes. The parking brakes shall be set, and the trailer subjected to a force sufficient to move the trailer. The wheels shall skid and not roll on dry, rough concrete.

4.6.20 Serviceability. The test stand shall be inspected and evaluated from the standpoint of ease of maintenance, servicing, and operation. Particular note shall be made with regard to provisions to prevent accumulation of dirt, snow, ice, et cetera, that may hinder servicing and operation.

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4.6.21 Final disassembly inspection. The test stand shall be disassembled to the maximum practical extent and a thorough examination of all parts conducted to determine if failure, distortion, excessive wear, or other unsatisfactory conditions exist. If measurements are made to determine wear, they shall be compared with similar measurements made during initial disassembly inspection. Test stands not conforming to requirements specified herein shall be rejected.

4.6.22 Differential pressure indicators. The high-pressure filter back-pressure indicators shall be tested for proper operation. The red light of the indicators shall come on when the differential pressure across the filters is 100 ±15 psi. The indicators shall operate at any line pressure between 750 and 5,000 psi.

4.6.23 Road test. The test stand shall be towed for a distance of 50 miles to determine general functioning of running gear components. The trailer shall be inspected for excessive vibrations, distortion, loosening, and maladjustment during and following this test. Evidence of malfunctioning shall be cause for rejection.

4.6.24 Maneuverability tests. The test stand shall be subjected to the following tests. During and following these tests, the test stand shall exhibit no evidence of failure or impending failure.

4.6.24.1 The test stand shall be towed 100 miles over improved roads at an average speed of 20 mph, maximum speed of 25 mph; and not less than 50 miles over graded road at an average speed of 10 mph, maximum speed of 12-1/2 mph.

4.6.24.1.1 Not less than 20 sudden stops shall be made while towing at each speed specified in 4.6.24.1.

4.6.24.2 The test stand shall make not less than 20 full turns to the left and 20 full turns to the right at 8 mph at its minimum turn radius.

4.6.25 Air transportability. The test stand shall be subjected to the air transportability tests specified in MIL-A-8421. During this test, the test stand shall be secured by means of the lifting and tiedown fittings only. Upon completion of this test, the test stand shall be inspected to determine adequacy of bracing, adequacy of tiedown and liftings, installation, and security of mounting of all components. The test stand will not be required to be operable upon completion of the 8-g acceleration loading. If it can be shown by computation that the test stand and all components will withstand the 8-g acceleration loading, the computation may be substituted in lieu of the test.

4.6.26 Reliability demonstration and test. Satisfactory completion of all tests specified herein demonstrates compliance with the quantitative reliability requirements of this specification.

4.6.27 Maintainability evaluation. The test stand shall be inspected and evaluated from the standpoint of ease of maintenance, servicing, and operation. Particular attention shall be given maintenance with a minimum number of common tools, servicing, and operation by personnel wearing heavy gloves, and provisions made to prevent accumulation of dirt, snow, ice, et cetera, that may hinder servicing and operation.

4.7 Inspection of preparation for delivery. The inspection of the preservation, packaging, packing, and marking shall be in accordance with the requirements of section 5 herein.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. The test stand shall be preserved and packaged in accordance with method I of MIL-P-116. Exterior, bare, or unpainted ferrous metal surfaces shall be coated with type P-1 preservative.

5.1.1.1 Hydraulic system. The hydraulic system shall be drained of operational fluid and filled to full operating capacity with preservative hydraulic fluid conforming to MIL-H-6083, type I. A tag containing the following information shall be attached to the hydraulic system in a conspicuous place:

CAUTION

Before placing test stand in operation, drain all preservative hydraulic fluid from the hydraulic system and refill to full operating capacity with hydraulic fluid conforming to MIL-H-5606.

All vent openings to the hydraulic system shall be covered with barrier material conforming to MIL-B-121, grade A, and sealed with tape conforming to PPP-T-60, class 1.

5.1.1.2 Compressor. The compressor shall be preserved and packaged as specified in MIL-C-3600.

5.1.1.3 Electric motor. The electric motor shall be preserved and packaged as specified in MIL-E-16298, alternate method.

5.1.1.4 Hose and electrical cable. Hose and electrical cable ends shall be wrapped with barrier material conforming to MIL-B-121, grade A, and sealed with tape conforming to PPP-T-60, class 1. They shall then be individually coiled to a minimum safe diameter and securely bound in three places with cotton tape. Coiled hose and electrical cable shall be placed in containers conforming to PPP-B-621, class 2, and securely fastened to test stand. Interior cushioning, blocking, and bracing for hose and cable shall be in accordance with MIL-STD-1186.

5.1.2 Level C. Preservation and packaging shall be in accordance with the manufacturer's commercial practice.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Each test stand preserved and packaged as specified in 5.1.1 shall be packed in accordance with MIL-STD-281, level A (mobile).

5.2.2 Level B. Each test stand preserved and packaged as specified in 5.1.1 shall be packed in accordance with MIL-STD-281, level B (mobile).

5.2.3 Level C. Each test stand preserved and packaged as specified in 5.1.2 shall be packed in accordance with MIL-STD-281, level C (mobile).

5.3 Marking. In addition to any special marking required by the contract or order, the unit packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The A/M27T-2 hydraulic test stand is intended for use in ground checking and maintenance of aircraft hydraulic systems.

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6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Fire and casualty hazards. The bidder will submit proof to the procuring activity that the electrical equipment and wiring specified in 3.9 that he proposes to supply under this specification conform to the standards of the Underwriters' Laboratories, Inc. as regards fire and casualty hazards. The label or listing of the Underwriters' Laboratories, Inc. will be accepted as evidence of conforming to the specification requirements.

In lieu of the label or listing, the bidder may submit independent proof satisfactory to the procuring activity that this wiring and electrical equipment conforms to the published standards, including methods of tests, of the Underwriters' Laboratories, Inc. Compliance with fire and casualty hazards requirements does not absolve the bidder from complete compliance with the other requirements of this specification.

c. Location and conditions for preproduction testing (see 4.4).

d. Level of preservation, packaging, and packing required (see section 5).

6.3 Definitions. For the purpose of this specification, the following definitions will apply:

6.3.1 Reliability. Reliability is defined as the probability of performing a specified function under given conditions without failure for a specified period of time. Recognizing that, in general, the rate of failure of equipment is fairly constant throughout the life of the equipment, the probability of nonfailure over an operating time interval decreases exponentially as a function of the length of the interval during which time there is a constant failure rate, and be expressed as follows:

Confidence level: 90 percent

$$\text{MTBF} = \frac{\text{Total test time}}{2.3}$$

$$\text{Reliability} = e^{-x} \text{ where } x = \frac{\text{Mission Time}}{\text{MTBF}}$$

The $2.3 = \frac{4.61}{2}$ is based on the constant

for the Poisson/Chi squared distribution, assuming an exponential (2 degrees freedom) distribution even though the failure rate for the test time is zero.

6.3.2 Mean-time-between-failures. The mean life or mean-time-between-failures is the arithmetical mean (average) of the operating time between failures.

6.3.3 Disassembled to the maximum practical extent is defined as disassembly of moving contacting surfaces which have been assembled by the manufacturer of the test stand and which are subject to wear (gears, bearings, control mechanisms, et cetera).

6.4 The high-pressure pumps on this test stand are mounted on standard accessory drive pads. Aircraft pumps can be substituted by the use of the following:

AN4139	Adapter-Type XVI Drive Flange
AN4182	Adapter-Type XVII Drive Spline

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6.5 International agreements. Certain provisions in 3.5, 3.8.1, 3.8.1.3, and 3.8.3 of this specification are the subject of international standardization agreement (American-British-Canadian Air Standardization Coordinating Committee Air Standard 11/14). Certain provisions in 1.1, 3.5, 3.8.1.3, 3.8.3, 3.8.19, and 3.21.2 of this specification are the subject of international standardization agreement (North Atlantic Treaty Organization Standardization Agreement 3510). When amendment, revision, or cancellation of this specification is proposed, the departmental custodians will inform their respective Departmental Standardization Offices so that appropriate action may be taken respecting the international agreement concerned.

* 6.6 The outside margins of this specification have been marked to indicate where changes (deletions, additions, et cetera) from the previous issue have been made. This has been done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in those notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Air Force - 11

Preparing activity:
Air Force - 11

Reviewer:
Air Force - 82

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).

SPECIFICATION

ORGANIZATION (Of submitter) | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

a. GIVE PARAGRAPH NUMBER AND WORDING.

b. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity) | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-8270

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